1.Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Ans.

2.List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Ans.

Device	Main Function	Real – Life Example
Modem	It can connects our home or office to the internet.	In home it receives the internet from our internet company..
Router	It can shares the internet with different devices.	It connect the our mobile, laptop, and smart TV to the internet.
Switch	It connects many device by using network cables.	In college lab it connect many computer together.
Access Point	In access point they provide Wi-Fi so wireless devices can connect to the network.	In the office they provides WiFi for employees laptop and phones.
Network Cable	This is use for to send data between devices.	A Ethernet cable connects a computer to a router or switch.

3.Take photos or find images online of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.
Ans.

4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Ans.

Feature	Router	Switch
Main use	Connects different networks, such as your home network to the Internet.	Connects multiple devices within the same network.
Typical location	Usually found at home, office, or near the Internet connection.	Usually found inside offices, computer labs, or server rooms.
How it handles data	Sends data between different networks using IP addresses.	Sends data to the correct device using MAC addresses.
Internet connection	Can provide access to the Internet.	Normally does not directly connect a network to the Internet.
Devices connected	Connects a local network to other networks.	Connects computers, printers, cameras, and other devices together.
Example	A home Wi-Fi router connects your phone and laptop to the Internet.	A switch connects several computers in a college computer lab.